

Data Communications

Introduction to Bandwidth and Throughput

Dr.S.Kanchana

Department of Computer Applications

Bandwidth

- Bandwidth refers to the maximum rate of data transfer across a given path or network connection.
- It is a measure of capacity, not speed.

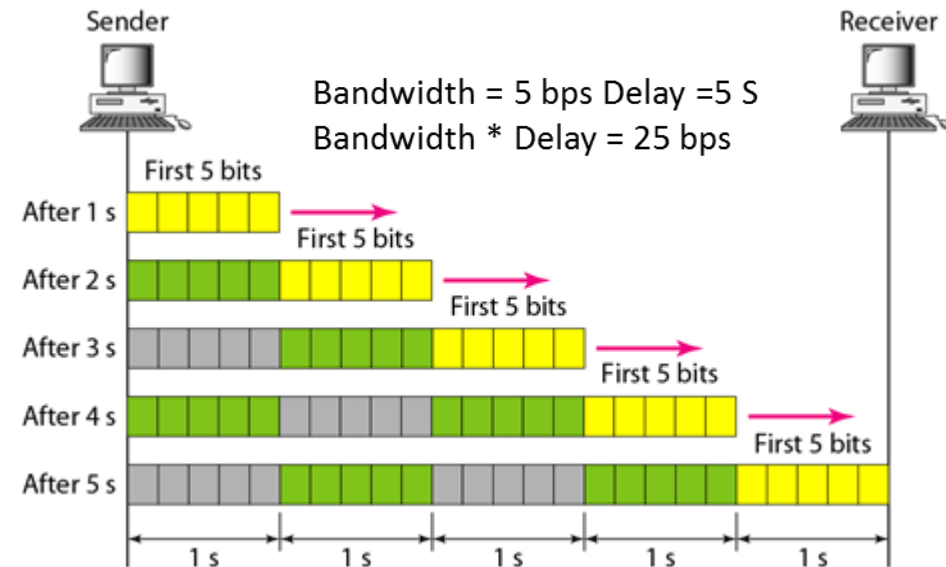
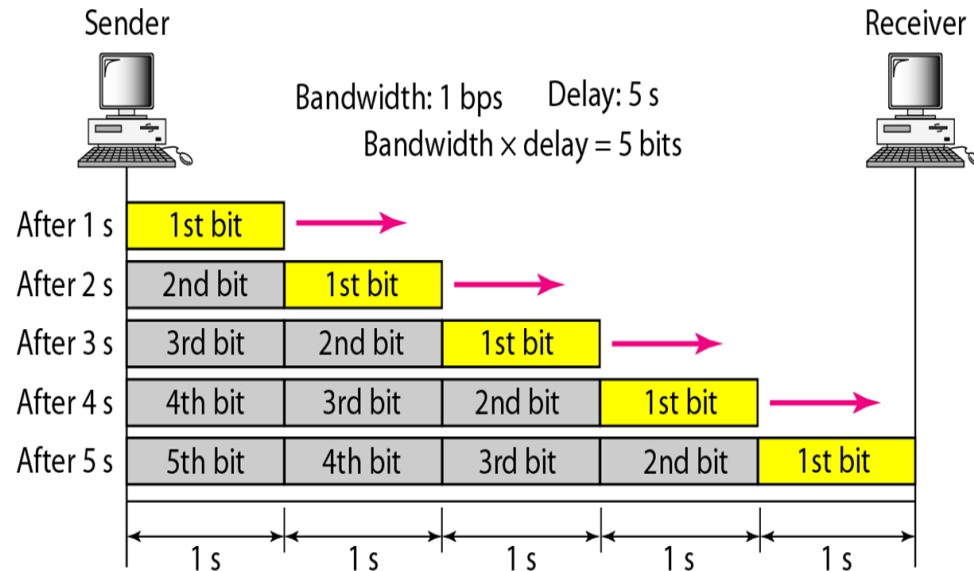
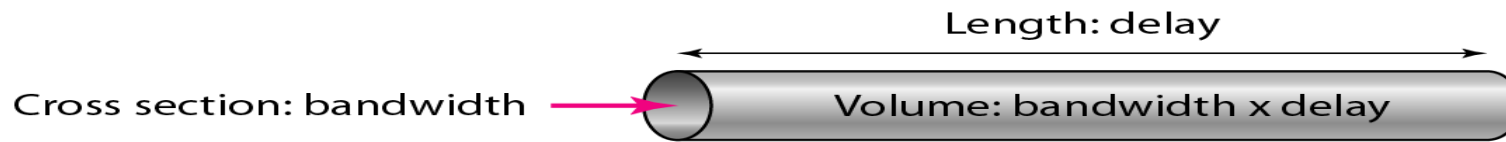
Example: Think of a network cable as a highway.

- Bandwidth is the number of lanes on the highway.
- A four-lane highway (high bandwidth) can handle more cars (data) simultaneously than a two-lane highway (low bandwidth).

Bandwidth is typically measured in:

- Bits per second (bps), with common multiples being:
- Mbps (Megabits per second): Millions of bits per second. (e.g., home internet plans)
- Gbps (Gigabits per second): Billions of bits per second. (e.g., fiber optic connections, data centers)

The bandwidth-delay product defines the number of bits that can fill the link.



Problem 1: Nyquist Formula (Noiseless Channel)

A noiseless channel has a bandwidth of 4 kHz and uses 16-level (M=16) signaling.

Calculate the maximum data rate.

$$C = 2 \times B \times \log_2(M)$$

$$C = 2 \times 4000 \times \log_2(16)$$

$$\log_2(16) = 4$$

$$C = 2 \times 4000 \times 4 = 32,000 \text{ bps or } 32 \text{ kbps}$$

If a channel transmits 2 Mbps using 4-level signaling ($M=4$) in a noiseless channel,
what is the minimum bandwidth required?

$$C=2 \times B \times \log_2(M)$$

$$2 \times B \times \log_2(4) = 2,000,000$$

$$\log_2(4) = 2$$

$$2 \times B \times 2 = 2,000,000$$

$$B = 2,000,000 / 4$$

$$= 500,000 \text{ Hz or } 500 \text{ kHz}$$

How long will it take to download a 2 GB file over a network connection with a bandwidth of 100 Mbps? (Assume no latency, errors, or other overhead)

Solution:

File Size = 2 GB = $2 * 2^{30}$ Bytes * 8 bits/Byte = 17,179,869,184 bits

Bandwidth = 100 Mbps = $100 * 10^6$ bps = 100,000,000 bps

Time = 17,179,869,184 bits / 100,000,000 bps = 171.8 seconds

Throughput

- Throughput is the actual rate at which data is successfully transmitted over a communication channel from sender to receiver during a specific period of time.
- It is usually measured in bits per second (bps), kilobits per second (kbps), or megabits per second (Mbps).
- It represents effective data transfer.
- Influenced by network congestion, protocol overhead, errors, and hardware limitations.
- $\text{Throughput} \leq \text{Bandwidth}$ (because bandwidth is the maximum possible rate, while throughput is the real achieved rate)

$$\text{Throughput} = \frac{\text{Total Data Transferred}}{\text{Time Taken}}$$

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Throughput – Numerical Problems



Example 1:

If 10 MB of data is sent in 8 seconds:

$$\begin{aligned}\text{Throughput} &= 10 \times 8,000,000 \text{ bits} / 8 \text{ sec} \\ &= 10,000,000 \text{ bps or } 10 \text{ Mbps}\end{aligned}$$

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Throughput – Numerical Problems

Example 2:

A file of 50 MB is downloaded in 40 seconds.

Find the throughput in Mbps.

Solution:

Convert MB to bits:

$50 \text{ MB} = 50 \times 8,000,000 = 400,000,000 \text{ bits}$

$\text{Throughput} = 400,000,000 \text{ bits} / 40 \text{ sec}$

$= 10,000,000 \text{ bps or } 10 \text{ Mbps}$

Example 3:

A network with bandwidth of 10 Mbps can pass only an average of 12,000 frames per minute with each frame carrying an average of 10,000 bits. What is the throughput of this network?

Solution

We can calculate the throughput as

$$\text{Throughput} = \frac{12,000 \times 10,000}{60} = 2 \text{ Mbps}$$

The throughput is almost one-fifth of the bandwidth in this case.

- **Transmission Medium**

Copper cables, fiber optics, and wireless all have different maximum bandwidths.

- **Signal-to-Noise Ratio (SNR)**

Higher noise reduces effective bandwidth.

- **Distance**

Longer distances can reduce bandwidth due to attenuation.

- **Multiplexing and Sharing**

Shared channels reduce available bandwidth per user.

- **Hardware Limitations**

Routers, switches, and NICs have maximum support

- **Network Congestion**

More traffic → less throughput per user.

- **Protocol Overhead**

Headers, acknowledgments, and control messages reduce effective data rate.

- **Error Rate and Retransmissions**

High error rates → more retransmissions → lower throughput.

- **Latency and Delay**

Higher delay reduces the rate at which data is acknowledged and sent.

- **Processing Power**

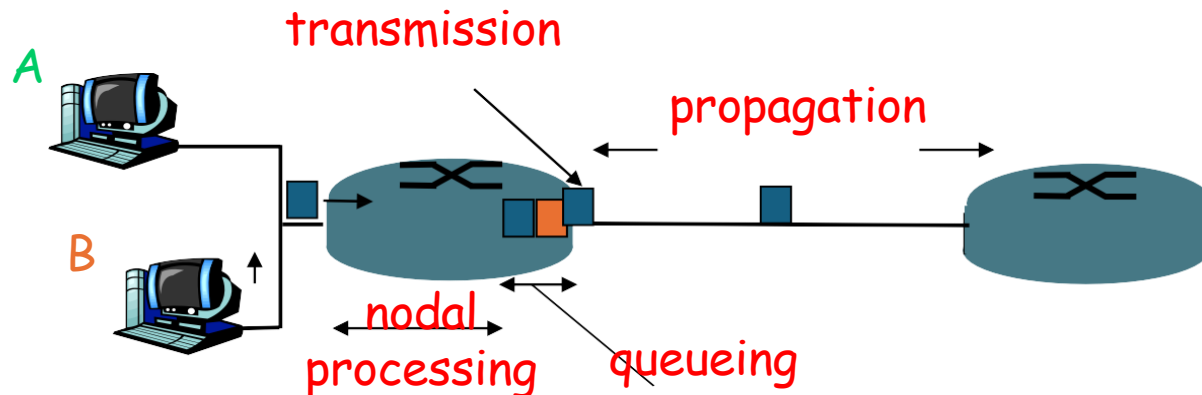
End devices and network equipment speed affect throughput.

Performance metrics in networks is delay, packet loss and throughput.

- Delay is the time taken to transmit the packet from sending host to receiving host

There are **Four Components**

- **Nodal Processing Delay** – Deciding the output Port and Checking for bit errors
- **Queuing Delay** – Waiting Time in the queue
- **Transmission Delay** – Time to transmit the file on to the link
- **Propagation Delay** – Time to carry the packet from one end of the link to another link



$$d_{\text{nodal}} = d_{\text{proc}} + d_{\text{queue}} + d_{\text{trans}} + d_{\text{prop}}$$

d_{proc} = processing delay

typically a few microsecs or less, It is negligible for high end machines

d_{queue} = queuing delay

depends on congestion, it is variable and depends on internet traffic

d_{trans} = transmission delay = L/R significant for low-speed links

d_{prop} = propagation delay = d/s

a few microsecs to hundreds of msecs, depends on distance between sender and receiver

As we all know in a network

- Processing delay is typically a few microsecs or less
(Today's processors are very fast)
- Queuing delay depends on congestion in the network
- Transmission delay depends on link bandwidth and
- Propagation delay depends on the distance between the two nodes

What is the propagation delay if the distance between the two points is 12,000 km? Assume the propagation speed to be 2.4×10^8 m/s in cable.

Solution

We can calculate the propagation delay as

$$t_p = \text{distance} / \text{propagation speed}$$

$$\text{Propagation time} = \frac{12,000 \times 1000}{2.4 \times 10^8} = 50 \text{ ms}$$

How long does it take to propagate a file of size 3.5KB over a link of distance 4000km.

The propagation speed is 2.5×10^8 m/sec and transmission rate is 6Mbps.

There is no processing delay and the data packets can be sent continuously (no queuing delay)?

$$\begin{aligned}\text{Propagation delay} &= d/s \\ &= 4000 \times 1000 / (2.5 \times 10^8) = 0.016 \text{ secs}\end{aligned}$$

$$\begin{aligned}\text{Transmission delay} &= L/R = (3.5 \times 1024 \times 8) / (6 \times 10^6) \\ &= 0.0047786 \text{ secs}\end{aligned}$$

Total Delay = Propagation Delay + Transmission Delay + Processing Delay + Queuing Delay

Total delay = 0.0207786 secs

How long does it take to propagate a file of size 1.5KB over a link of distance 5000km. The propagation speed is 2.5×10^8 m/sec and transmission rate is 10Mbps. There is no processing delay and the data packets can be sent continuously (no queuing delay)?

$$\begin{aligned}\text{Propagation delay} &= d/s \\ &= 5000 \times 1000 / (2.5 \times 10^8) \\ &= 0.02 \text{ secs}\end{aligned}$$

$$\begin{aligned}\text{Transmission delay} &= L/R \\ &= (1.5 \times 10^3 \times 8) / (10 \times 10^6) \\ &= 0.0012 \text{ secs}\end{aligned}$$

Total Delay = Propagation Delay + Transmission Delay + Processing Delay + Queuing Delay

Total delay = 0.0212 secs



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S Kanchana

Department of Computer Applications

kanchana@pes.edu

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Bandwidth Utilization concepts

Dr.S.Kanchana

Department of Computer Applications

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Efficient use of Bandwidth

Dr.S.Kanchana

Department of Computer Applications

Efficient use of Bandwidth

Efficient use of bandwidth in data communication:

- optimizing the data transfer so that the available channel capacity is utilized fully
- reducing overhead
- minimizing idle times
- preventing unnecessary retransmissions.

Bandwidth Efficiency= Data Rate(bps) / Bandwidth (Hz)

Higher bandwidth efficiency means better utilization of the channel.

Efficient use of Bandwidth

Multiplexing: FDM (Frequency Division), TDM (Time Division), WDM (Wavelength Division) for channel sharing.

- Compression: Reduces data size before transmission (e.g., JPEG, MP4, ZIP).
- Efficient Encoding: Use schemes like QAM, PCM to transmit more bits per signal change.
- Error Detection and Correction: FEC to reduce retransmissions and save bandwidth.
- Protocol Optimization: Reduce header overhead (e.g., UDP for real-time streaming).
- Traffic Shaping and Congestion Control: Avoid packet loss and retransmission for better flow. Bandwidth Efficiency

Bandwidth Efficiency= Data Rate(bps) / Bandwidth (Hz)

Higher bandwidth efficiency means better utilization of the channel.

Example:

If a system transmits 2 Mbps using a bandwidth of 500 kHz:

Efficiency=2,000,000/500,000=4 bps/Hz

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Bit rate & Baud Rate

Dr.S.Kanchana

Department of Computer Applications

Bit Rate

Bit rate (or data rate) is the number of bits transmitted per second over a communication channel. It represents how many data bits are sent in one second.

Unit: bits per second (bps), kbps, Mbps, Gbps.

Formula:

Bit Rate = Baud Rate $\times \log_2(M)$, where M = number of signal levels

Bit Rate

Example:

If a system transmits 1 bit per signal change and the signaling rate is 1,200 baud,

$$\text{Bit Rate} = 1,200 \times 1 = 1,200 \text{ bps}$$

If 4 levels ($M = 4$) are used:

$$\text{Bit Rate} = 1,200 \times \log_2(4) = 1,200 \times 2 = 2,400 \text{ bps}$$

Baud Rate

Baud rate is the number of signal units (symbols) transmitted per second. Each signal unit may carry one or more bits, depending on the modulation technique.

Unit: baud (symbols per second).

Example:

If the baud rate is 2,400 baud and each symbol carries 3 bits

$$\text{Bit rate} = 2400 * 3 = 7,200 \text{ bps}$$



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S Kanchana

Department of Computer Applications

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Multiplexing

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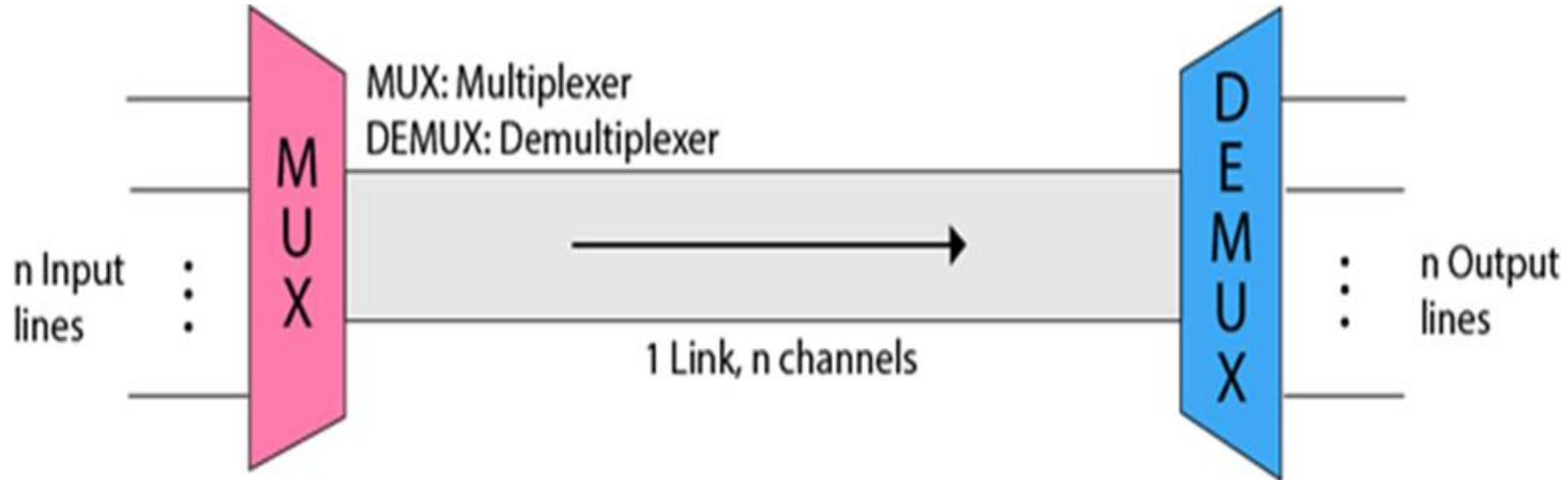
Multiplexing

- Multiplexing is a technique used to combine and send the multiple data streams over a single medium. The process of combining the data streams is known as multiplexing.
- Multiplexing is achieved by using a device called Multiplexer (**MUX**) that combines n input lines to generate a single output line.
- Demultiplexing is achieved by using a device called Demultiplexer (**DEMUX**) available at the receiving end.
- DEMUX separates a signal into its component signals (one input and n outputs).

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Multiplexing

Multiplexing



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Multiplexing

Types of Multiplexing

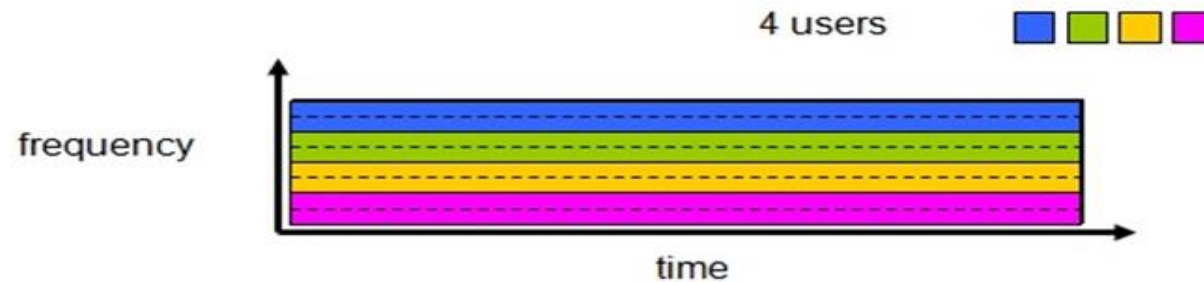
- Frequency division multiplexing
- Time division multiplexing
- Statistical division multiplexing

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Multiplexing

Frequency Division Multiplexing (FDM)

- FDM deals with analog signals.
- The available bandwidth of a single transmission medium is subdivided into several channels. The input signals are translated into frequency bands by using modulation techniques, and they are combined by a multiplexer to form a composite signal.



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Multiplexing

Frequency Division Multiplexing (FDM)

- The main aim of the FDM is to subdivide the available bandwidth into different frequency channels and allocate them to different devices
- Each signal is modulated onto a different carrier frequency and carrier frequencies are separated by ***guard bands***.
- The bandwidth of the transmission medium exceeds the required bandwidth of all the signals.

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Multiplexing

Advantages of FDM

- The process is simple and easy to modulate.
- A corresponding multiplexer or de-multiplexer is on the end of the high-speed line and separates the multiplexed signals.
- Analog signaling is used in order to transmit the signals.

Disadvantages of FDM

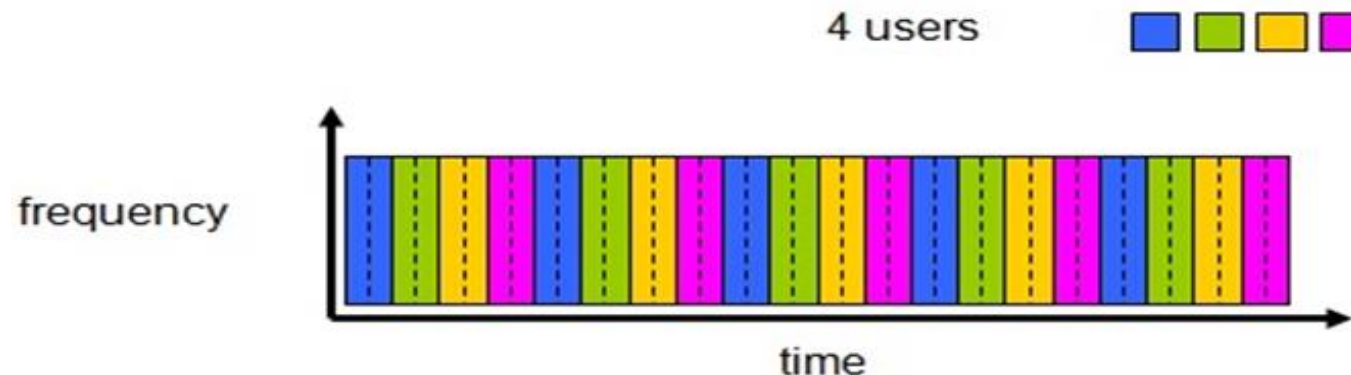
- It suffers the problem of crosstalk.
- A large number of modulators are required.
- It requires a high bandwidth channel.

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Multiplexing

Time Division Multiplexing(TDM)

- It is a digital technique.
- In **Time Division Multiplexing technique**, the total time available in the channel is distributed among different users. So, each user is allocated with different time interval known as a ***Time slot*** at which data is to be transmitted by the sender.



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Time Division Multiplexing(TDM)

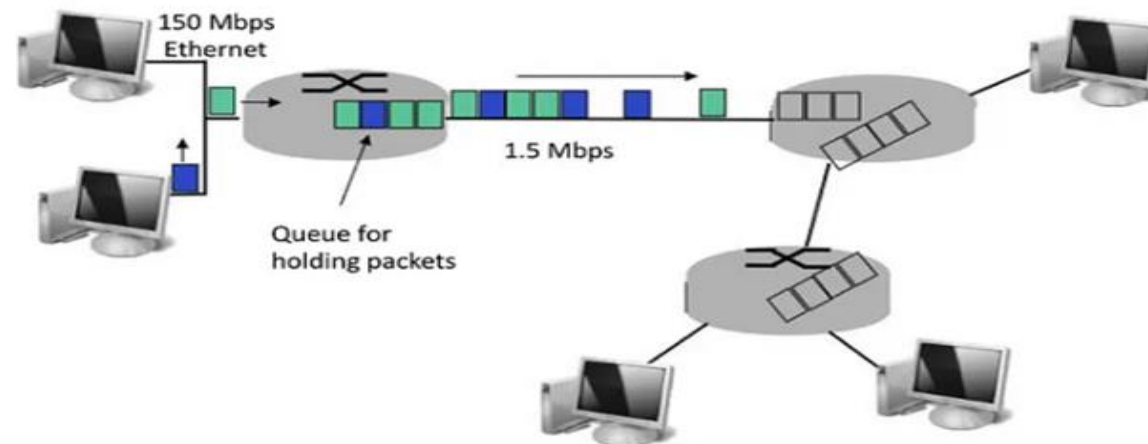
- A user takes control of the channel for a fixed amount of time.
- In TDM technique, data is not transmitted simultaneously rather the data is transmitted one-by-one.
- The signal is transmitted in the form of frames. Frames contain a cycle of time slots in which each frame contains one or more slots dedicated to each user.
- The main drawback of TDM is if there is no information to communicate after sometime, then the time slot gets wasted.

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Multiplexing

Statistical Multiplexing(SDM)

Statistical multiplexing is implemented to overcome the drawback of TDM and FDM. Here like TDM, the time is divided into n time slots and assigned to each channel **on demand** ie, related to workload or priority and also determines how much time each user should get. This is more efficient technique.





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S Kanchana

Department of Computer Applications

kanchana@pes.edu

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Introduction To Data Link Layer



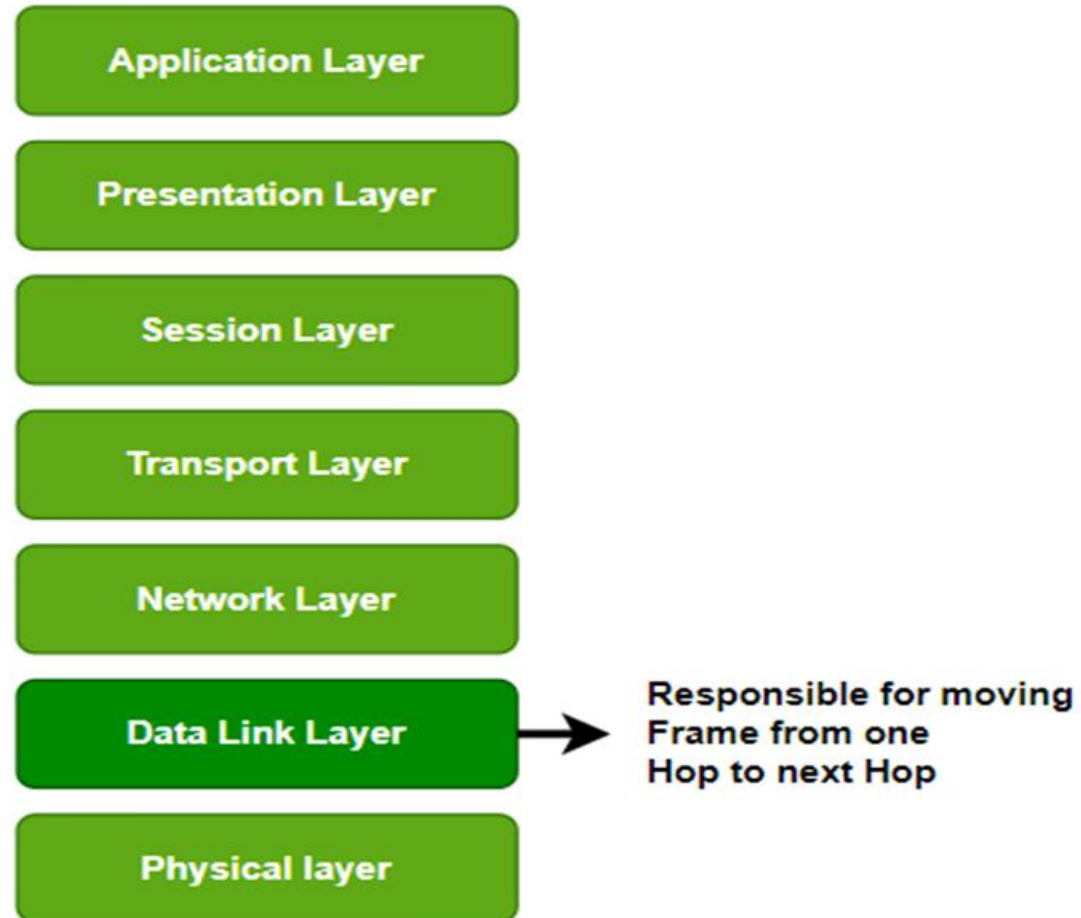
Introduction

- The data link layer is the second layer from the bottom in the OSI network architecture model.
- It is responsible for the node-to-node delivery of data.
- Its major role is to ensure error-free transmission of information.
- DLL is also responsible for encoding, decoding, and organizing the outgoing and incoming data.

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Data Link Layer

DLL in OSI



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Data Link Layer - Sub-Layers of The Data Link Layer

Logical Link Control (LLC)

- LLC deals with multiplexing, the flow of data among other applications
- LLC is responsible for providing error messages and acknowledgments.

Media Access Control (MAC)

- MAC sub layer manages the device's interaction, responsible for addressing frames, and controls physical media access.

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Functions of Data Link Layer

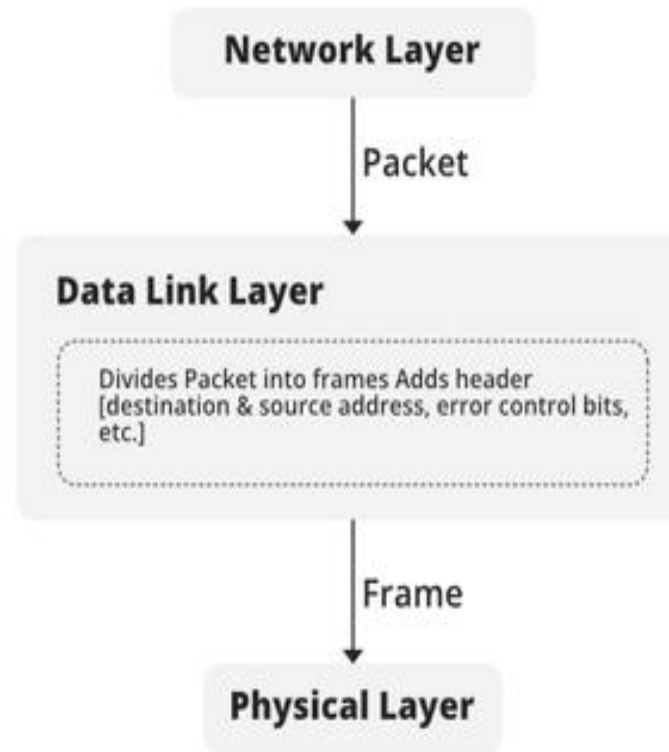
Framing

- The packet received from the Network layer is known as a frame in the DLL.
- At the sender's side, DLL receives packets from the Network layer and divides them into small frames, then, sends each frame bit-by-bit to the physical layer.
- It attaches some special bits for error control and addressing at the header and end of the frame.
- At the receiver's end, DLL takes bits from the Physical layer organizes them into the frame, and sends them to the Network layer.

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Functions of Data Link Layer

Framing



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Functions of Data Link Layer

Addressing

- The data link layer encapsulates the source and destination's MAC address/physical address in the header of each frame.
- MAC address is the unique hardware address that is assigned to the device while manufacturing.

Error Control

- Data can get corrupted due to various reasons like noise, attenuation, etc.
- It is the responsibility of the data link layer, to detect the error in the transmitted data and correct it using error detection and error correction techniques.

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Functions of Data Link Layer



Flow Control

If the receiver's receiving speed is lower than the sender's sending speed, this can lead to an overflow in the receiver's buffer and some frames may get lost.

So, DLL synchronize the sender's and receiver's speeds and establish flow control between them.

Access Control

When multiple devices share the same communication channel there is a high probability of collision, so it's the responsibility of DLL to check which device has control over the channel and CSMA/CD and CSMA/CA can be used to avoid collisions and loss of frames in the channel



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Link Layer Addressing



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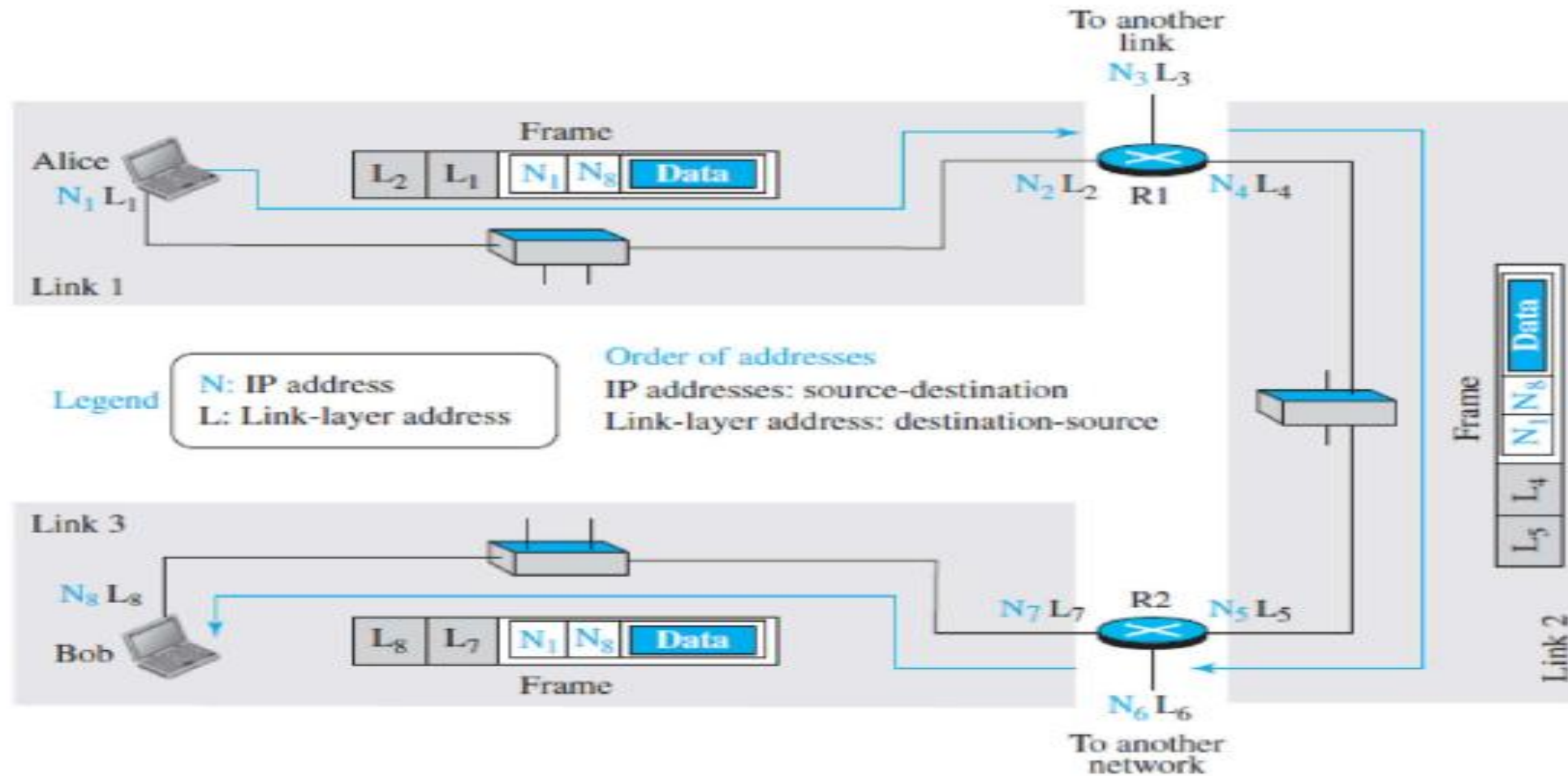
Link Layer Addressing

Introduction

- A link-layer address is called a link address or physical address or MAC address.
- Since a link is controlled at the data-link layer, the addresses need to belong to the data-link layer.
- When a datagram passes from the network layer to the data-link layer, the datagram will be encapsulated in a frame and two data-link addresses are added to the frame header. These two addresses are changed every time the frame moves from one link to another.

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Link Layer Addressing



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Link Layer Addressing

Consider we have two hosts: Alice (source) and Bob (destination).

- For each host, there is two addresses, the IP addresses (N) and the link-layer addresses (L).
- We have three frames, one in each link. Each frame carries the same datagram with the same source and destination addresses (N1 and N8), but the link-layer addresses of the frame change from link to link.
- In link 1, the link-layer addresses are L1 and L2.
- In link 2, they are L4 and L5.
- In link 3, they are L7 and L8.

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Link Layer Addressing

- The IP addresses and the link-layer addresses are not in the same order.
- For IP addresses, the source address comes before the destination address; for link-layer addresses, the destination address comes before the source.

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Link Layer Addressing - Three types of Addresses

Unicast Address

- Each host or each interface of a router is assigned a unicast address.
- Unicasting means **one to-one** communication.
- A frame with a unicast address destination is destined only for one entity in the link.

Example

The unicast addresses are 48 bits (six bytes) that are presented as 12 hexadecimal digits separated by colons. Ex: a link-layer address of a computer. A2:34:45:11:92:F1

First byte = A2

A3 in binary 1010 0010

LSB = 0 (unicast)

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Link Layer Addressing - Three types of Addresses

Multicast Address

- Some link-layer protocols define multicast addresses.
- Multicasting means ***one-to many*** communication.

Example

Example for a multicast address: A3:34:45:11:92:F1

First byte = A3

A3 in binary 1010 0011

LSB = 1 (Multicast)

If the binary ends with 0, then it is unicast and

If the binary ends with 1, then it is multicast

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Link Layer Addressing - Three types of Addresses

Broadcast Address

- Some link-layer protocols define a broadcast address.
- Broadcasting means ***one-to-all*** communication.
- A frame with a destination broadcast address is sent to all entities in the link.

Example

The broadcast link-layer addresses in LAN, Ethernet, are 48 bits, all 1s, that are presented as 12 hexadecimal digits separated by colons.

Example for a broadcast address: FF:FF:FF:FF:FF:FF



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Address Resolution Protocol



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Address Resolution Protocol

- The ARP is one of the protocols defined in the network layer.
- It maps an IP address to a logical-link address.
- The main work of ARP : It accepts an IP address from the IP protocol, maps the address to the corresponding link layer address, and passes it to the data-link layer.



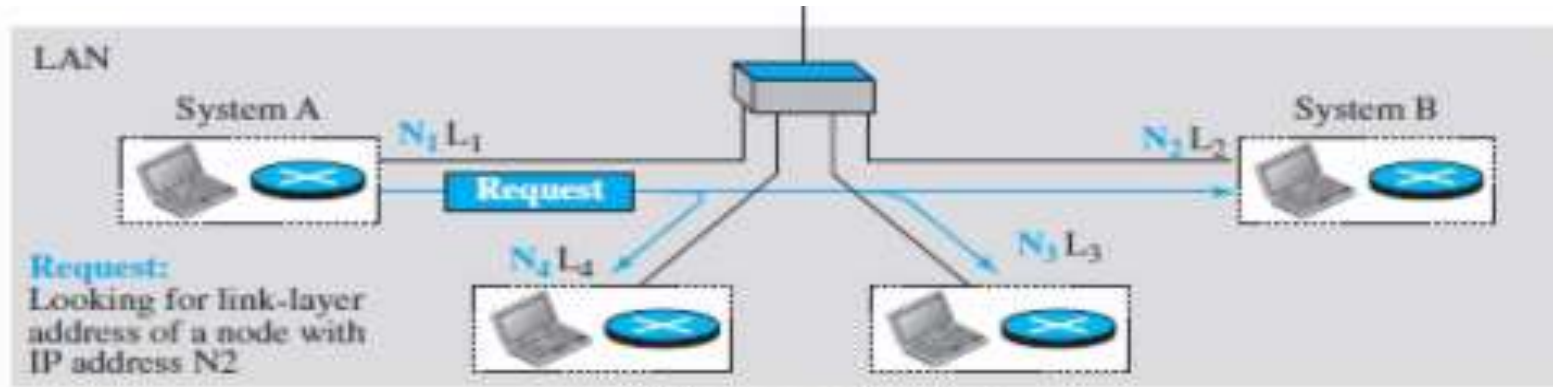
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Address Resolution Protocol

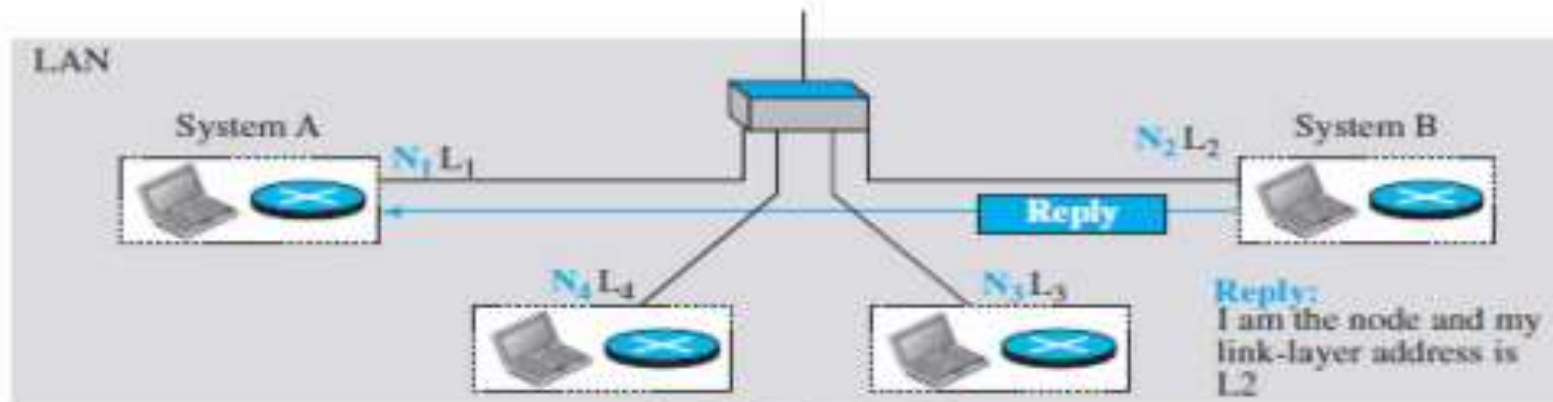
- If a host or a router needs to find the link-layer address of another host or router in its network, it sends an ARP request packet.
- The packet includes the link-layer and IP addresses of the sender and the IP address of the receiver.
- As the sender does not know the link-layer address of the receiver, the query is broadcast over the link using the link-layer broadcast address.

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Address Resolution Protocol



a. ARP request is broadcast



b. ARP reply is unicast

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Address Resolution Protocol

Working of ARP

- Every host or router on the network receives and processes the ARP request packet, but only the intended recipient recognizes its IP address and sends back an ARP response packet.
- The system on the left (A) has a packet that needs to be delivered to another system (B) with IP address N2.
- System A needs to pass the packet to its data-link layer for the actual delivery, but it does not know the physical address of the recipient.

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Address Resolution Protocol

Working of ARP(cont'd)

- It uses the services of ARP, to send a **broadcast ARP request** packet to ask for the physical address of a system with an IP address of N2. This packet is received by every system on the physical network, but only system B will answer it.
- System B sends an **unicast ARP reply packet** that includes its physical address.
- Now system A can send all the packets it has for this destination using the physical address it received.



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Standard Ethernet



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Standard Ethernet



Introduction

The original Ethernet was created in 1976 at Xerox's Palo Alto Research Center (PARC). Since then, it has gone through four generations:

- **Standard Ethernet (10 Mbps),**
- **Fast Ethernet (100 Mbps),**
- **Gigabit Ethernet (1 Gbps)**
- **Ten-Gigabit Ethernet (10 Gbps)**

The Ethernet technology with the data rate of 10 Mbps is referred to as the Standard Ethernet.

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Standard Ethernet

Characteristics of Standard Ethernet

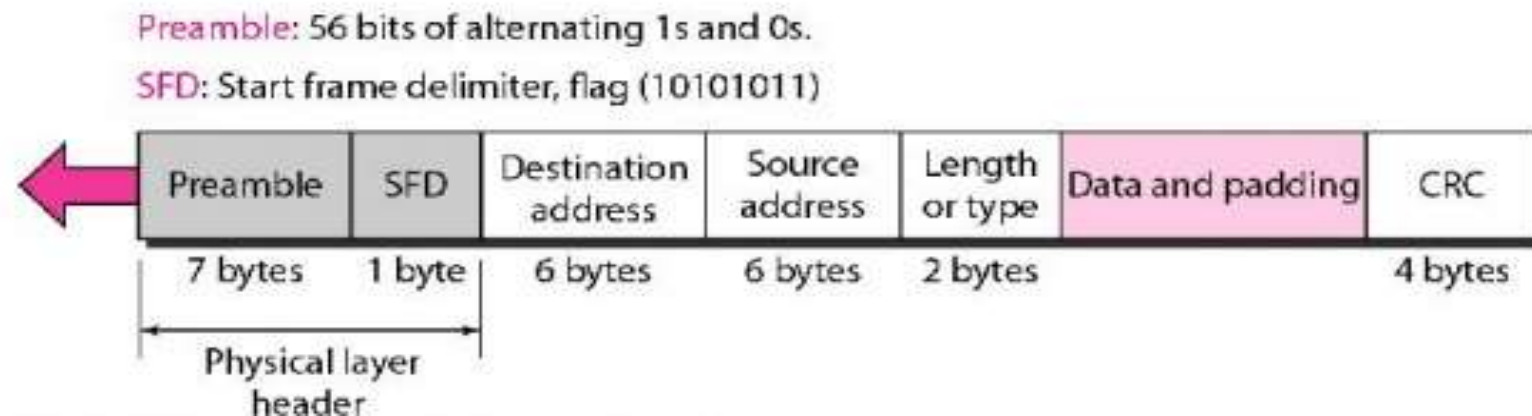
- Ethernet provides connectionless service. Each frame sent is independent of the previous or next frame.
- Ethernet has no connection establishment or connection termination phases.
- Ethernet is unreliable

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Standard Ethernet

Frame Format

The Ethernet frame contains seven fields: preamble, SFD, DA, SA, length or type of protocol data unit (PDU), upper-layer data, and the CRC. Ethernet does not provide any mechanism for acknowledging received frames. Acknowledgments must be implemented at the higher layers.



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Standard Ethernet



Cont'd

Preamble: This field contains 7 bytes (56 bits) of alternating 0s and 1s that alerts the receiving system to the coming frame and enables it to synchronize its input timing. The pattern provides only an alert and a timing pulse.

Start frame delimiter (SFD): The second field (1 byte: 10101011) signals the beginning of the frame. The last 2 bits are 11 and alerts the receiver that the next field is the destination address.

Destination address (DA): The DA field is 6 bytes and contains the physical address of the destination station or stations to receive the packet.

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Standard Ethernet



Cont'd

Source address (SA): The SA field is also 6 bytes and contains the physical address of the sender of the packet.

Length or type: This field define the upper-layer protocol whose packet is encapsulated in the frame.

Data: This field carries data encapsulated from the upper-layer protocols. It is a minimum of 46 and a maximum of 1500 bytes.

CRC: The last field contains error detection information.

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Standard Ethernet

Frame Length

- Ethernet has restrictions on both the minimum and maximum lengths of a frame.
- An Ethernet frame needs to have a minimum length of 512 bits or 64 bytes.
- The standard defines the maximum length of a frame as 1518 bytes.
- Minimum data length is 46 bytes
- Maximum data length is 1500 bytes

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Standard Ethernet

Addressing

Each station on an Ethernet network has its own network interface card (NIC).

The NIC provides the station with a 6-byte physical address.

The Ethernet address is 6 bytes (48 bits), nominally written in hexadecimal notation, with a colon between the bytes.

06 : 01 : 02 : 01 : 2C : 4B

└──────────┘ └──────────┘ └──────────┘
6 bytes 12 hex digits 48 bits

Fig: example for an Ethernet address in hexadecimal notation

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Standard Ethernet - Addressing



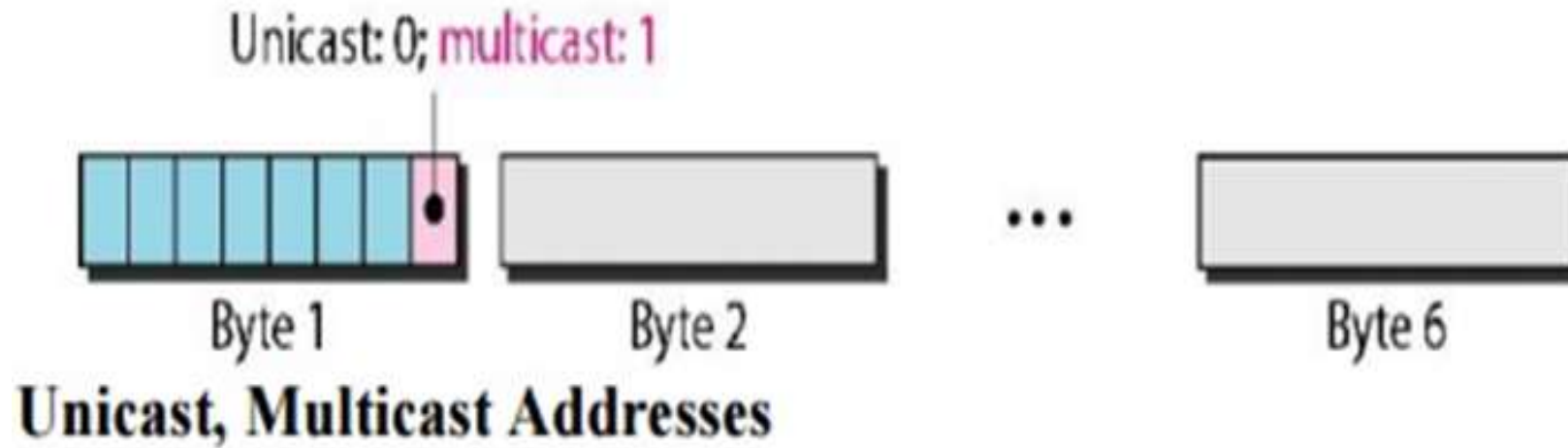
Unicast, Multicast, and Broadcast Addresses:

- A source address is always a unicast address-the frame comes from only one station.
- The destination address can be unicast, multicast, or broadcast.
- If the least significant bit of the first byte in a destination address is 0, the address is unicast; otherwise, it is multicast.
- The broadcast address is a special case of the multicast address; the recipients are all the stations on the LAN.
- A broadcast destination address is forty-eight 1s.

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Standard Ethernet - Addressing

Unicast, Multicast, and Broadcast Addresses



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Standard Ethernet - Addressing

Define the type of the following destination addresses:

- a. 4A:30:10:21:10:1A
- b. 47:20:1B:2E:08:EE
- c. FF:FF:FF:FF:FF:FF

Solution

To find the type of the address, look at the second hexadecimal digit from the left. If it is even, the address is unicast. If it is odd, the address is multicast. If all digits are F's, the address is broadcast. Therefore the result is the following:

- This is a unicast address because A in binary is 1010 (even).
- This is a multicast address because 7 in binary is 0111 (odd).
- This is a broadcast address because all digits are F's



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Media Access Control



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Media Access Control

Introduction

- A sub layer of the Data Link Layer is Medium Access Control (MAC).
- Its main duty is to control access to a network's shared communication medium, ensuring that various devices can transmit data effectively and fairly over a common communication path like a wired Ethernet .

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Media Access Control

- When a sender and receiver have a dedicated link to transmit data packets, the data link control is enough to handle the channel.
- If there is no dedicated path between two devices, multiple stations access the channel and simultaneously transmits the data over the channel. It may create collision and cross talk.
- Thus, the multiple access protocol is required to reduce the collision and avoid crosstalk between the channels.

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Media Access Control

The different multiple access protocols are

- **CSMA**
- **CSMA /CD**
- **CSMA / CA**

DATA COMMUNICATIONS

Media Access Control

CSMA/ CD - Carrier sense multiple access/ collision detection

- The CSMA/CD protocol works with a medium access control layer.
- It first senses the shared channel before broadcasting the frames, and if the channel is idle, it transmits a frame to check whether the transmission was successful.
- If the frame is successfully received, the station sends another frame.
- If any collision is detected in the CSMA/CD, the station sends a ***jam/ stop signal*** to the shared channel to terminate data transmission.
- After that, it waits for a random time before sending a frame to a channel.



THANK YOU

Sumitha Chandran
Department of Computer Applications

Data Communications

Wireless LAN-802.11



Introduction

- A **Wireless Local Area Network (WLAN)** is a type of computer network that allows devices (laptops, smartphones, printers, etc.) to connect and communicate **without physical cables**.
- It uses **radio waves (Wi-Fi technology)** to provide mobility and flexibility.
- WLAN is commonly used in **homes, offices, colleges, airports, public hotspots, etc.,**

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Wireless LAN - Features

- Wireless Communication** – No need for cables, devices connect via radio frequency (Wi-Fi).
- Mobility** – Users can move around within the network coverage area while staying connected.
- Scalability** – Easy to add more devices without extra wiring.
- Flexibility** – Supports various devices like laptops, smartphones, and tablets.
- Cost Effective** – Saves cost of physical cabling and installation.
- Roaming Support** – Users can roam between access points without losing connection.
- High Speed** – Provides data transfer speeds ranging from a few Mbps to several Gbps (latest Wi-Fi 6/6E).
- Security Features** – Supports encryption and authentication (WPA, WPA2, WPA3).

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Wireless LAN - Architecture

WLAN architecture is mainly divided into **two types**

- **Basic Service Set (BSS)**
- **Extended Service Set (ESS)**

DATA COMMUNICATIONS

Wireless LAN - Architecture

Basic Service Set (BSS)

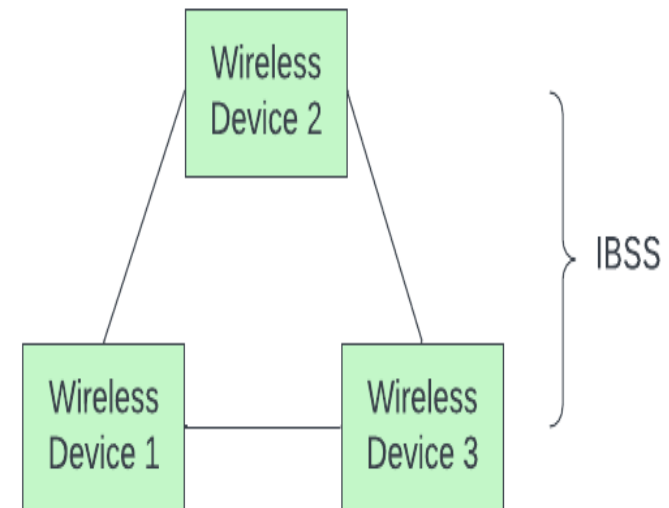
BSS is the smallest unit of a WLAN. It consists of:

- **Stations (STAs):** Stations comprise all devices and equipment that are connected to the wireless LAN. Devices like laptops, smartphones, printers.
- **Access Point (AP):** Works as a hub or central controller or wireless routers that bridge connections for base stations.

Two types of BSS: - Independent BSS, Infrastructure BSS

a) Independent BSS (IBSS)

- Also called **Ad-hoc network**.
- Devices communicate **directly with each other** (peer-to-peer).
- No Access Point is used.
- Example: Two laptops sharing files over Wi-Fi directly.

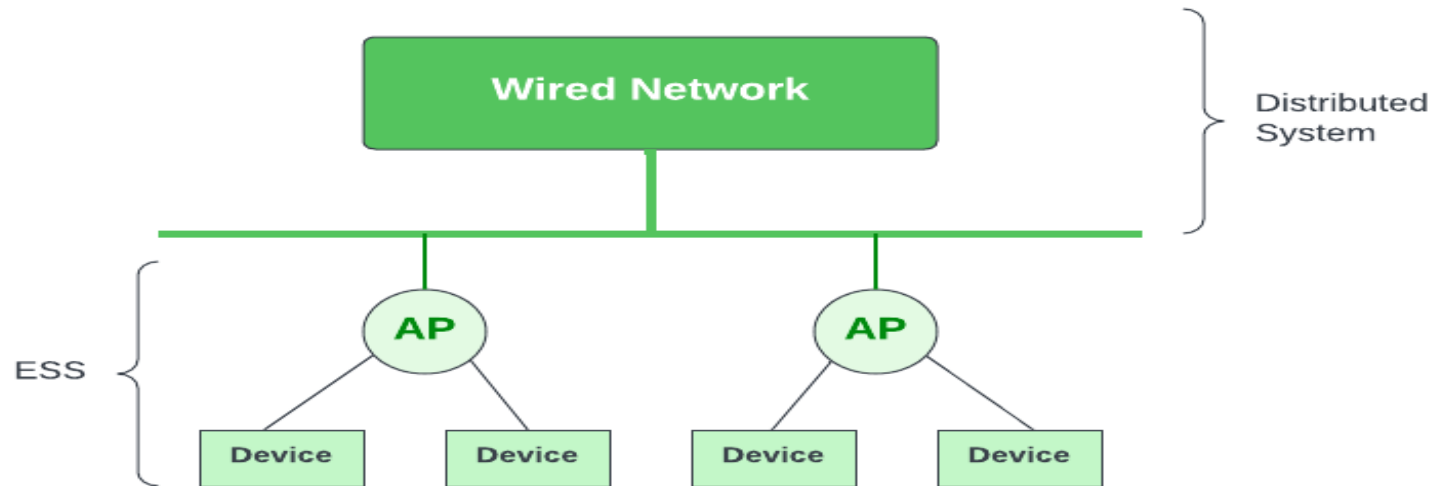


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Wireless LAN - Architecture

b) Infrastructure BSS

- Most commonly used form.
- Devices communicate **through an AP**.
- AP also connects the wireless network to the wired LAN or the internet.
- Example: Home Wi-Fi router connecting phones/laptops to the internet.

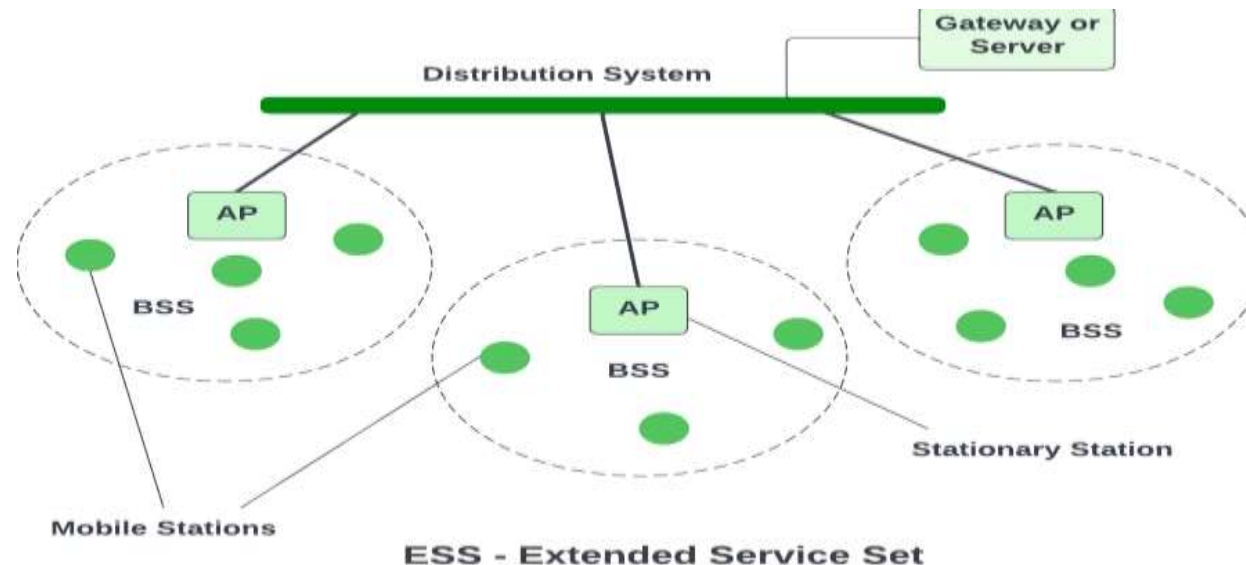


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Wireless LAN - Architecture

Extended Service Set (ESS)

- An ESS is formed by the combination of multiple BSSs connected together.
- Each BSS has its own Access Point, and they are connected through a **distribution system**.
- Provides **larger coverage area** and **roaming capability** (e.g., entire campus Wi-Fi).
- Example: A college Wi-Fi system where a student can walk across classrooms, library, and labs, and still stay connected.



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Wireless LAN - Addressing

- In a Wireless LAN (Wi-Fi network), every device (laptop, mobile, access point, printer, etc.) have a way to be **uniquely identified** so that data can be sent to the right device.
- This identification is done using **MAC addresses**. Each device in the WLAN has a unique MAC address.

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Wireless LAN – Frame format of WLAN

Frame Control - It is 2 bytes long and defines type of frame and control information.

Duration / ID - It contains the value indicating the period of time in which the medium is occupied (in μs).

Address 1 to 4 - It contains MAC addresses (48 bit each).

SC (Sequence Control) - It consists of 2 sub-fields i.e. sequence number (12 bits) and fragment number (4 bits).

Data - It contains information which is transferred transparently from a sender to the receiver.

CRC - It contains 32 bit CRC error detection sequence to ensure error free frame

Frame control	Duration /ID	Address 1	Address 2	Address 3	SC	Address 4	Data	CRC
2 bytes	2 bytes	6 bytes	6 bytes	6 bytes	2 bytes	6 bytes	0 - 2312 bytes	4 bytes

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Wireless LAN - Addressing

Types of WLAN Addresses

In Wireless LAN, there can be **up to 4 addresses** inside a frame. This is because data may need to pass through **Access Points** and **Distribution Systems** before reaching the destination.

Source Address (SA): The device that originally sends the data.

Destination Address (DA): The final receiving device.

Transmitter Address (TA): The device currently transmitting the frame (e.g., the Access Point).

Receiver Address (RA): The immediate device receiving the frame.



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Sumitha Chandran
Department of Computer Applications

Data Communications

Framing Techniques



DATA COMMUNICATIONS

Framing Techniques - Introduction

Framing is an important aspect of data link layer protocol design because it allows the transmission of data to be organized and controlled, ensuring that the data is delivered accurately and efficiently.

Framing means **breaking the stream of bits from the network layer into manageable units called frames.**

Different **framing techniques** are used to separate one frame from another.

The main techniques are:

- Character Count**

- Byte Stuffing**

- Bit Stuffing**

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Framing Techniques – Character Count

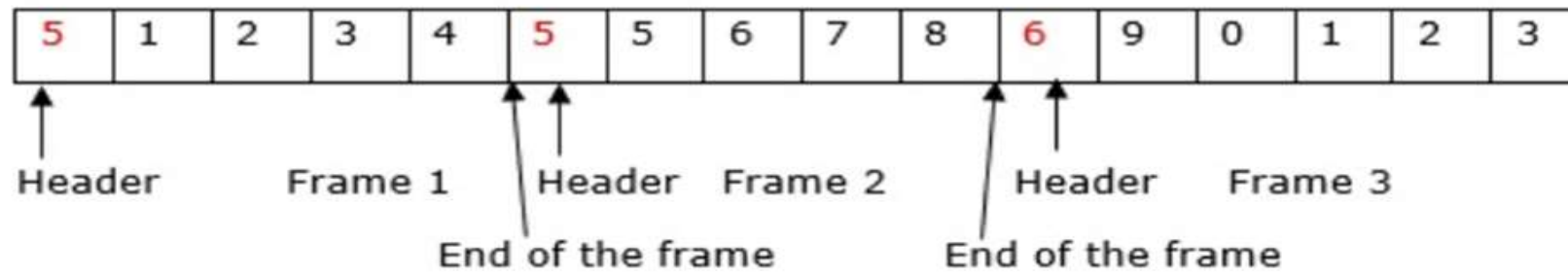
Character Count

- First framing method
- Each frame begins with a **field (header)** that specifies the **number of characters (bytes)** in that frame.
- When the data link layer at the destination sees the character count, it knows how many characters follow and hence where the end of the frame is.
- This helps the receiver know where one frame ends and the next frame starts.

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Framing Techniques – Character Count

Example - Consider a data – 1 2 3 4 5 6 7 8 9 0 1 2 3. Divide this into 3 frames and a header field is added to the beginning of each frame.



Suppose we want to send three frames:

Frame 1: "HELLO" (5 characters)

Frame 2: "WORLD" (5 characters)

Frame 3: "DATA" (4 characters)

They will be transmitted as: **[5 HELLO] [5 WORLD] [4 DATA]**

Disadvantage:

If the **count field gets corrupted** (e.g., wrong value), the receiver may lose synchronization and read the wrong frame.

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Framing Techniques – Byte Stuffing

Byte Stuffing (Character stuffing)

- Special **flag bytes** are used to mark the **start and end of a frame**.
- A common flag byte is: **DLE STX** (start) and **DLE ETX** (end) is used
- A **special flag byte** is used to mark the **start and end of a frame** (example: FLAG = F).
- If the **flag byte appears in the data**, we insert an **escape byte (E)** before it.
- If the **escape byte (E)** itself appears in the data, we again stuff an extra E before it.

Example: Data = A B F C E D

When sender sees F inside data, it adds E before it → E F

When sender sees E inside data, it adds another E before it → E E

So stuffed data becomes: **A B E F C E E D**

Transmitted Frame: **F A B E F C E E D F**

DATA COMMUNICATIONS

Framing Techniques – Byte Stuffing

Byte Stuffing (cont'd)

Receiver Side (De-stuffing):

- Receiver removes start and end F.
- Then scans inside data:
 - Whenever it sees E F, it removes E and takes F as data.
 - Whenever it sees E E, it removes one E and takes the second E as data.
- So final received data = **A B F C E D**

Disadvantage:

Extra bytes increases the over head

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Framing Techniques – Bit Stuffing

- Works at **bit level** (not byte).
- A special **flag pattern** is used to mark start and end of frame. Common flag: 01111110
- To ensure that this flag does not appear in the data, whenever the sender's data has **five consecutive 1s**, it **inserts a 0** (this is "stuffing").
- At the receiver side, after seeing five 1s, if the next bit is 0, it is removed.

Example:

Flag = 01111110

Data to send: 01111110 (contains the same pattern as flag)

Stuffing:- After every sequence of 5 ones (11111), insert a 0.

So, actual transmitted data: **011111010**

Full transmitted frame: **01111110 011111010 01111110**

(Start Flag + Stuffed Data + End Flag)

Receiver removes the stuffed 0 and retrieves the original data.

The main disadvantage is **more processing is required** for bit stuffing



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Sumitha Chandran
Department of Computer Applications